COMPASS_FE_20221220

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R Markdown

#notes I don't have a system yet to compare old slopes to current slopes for that day #I don't have a system for eliminating wells if the sd is high - if you think one of your sample wells was off you would need to remove it manually…

#Files Required: #csv of standards as list not plate with columns: Fe(II), Abs, Curve, FE ; these are: Known std concentration, Absorbance, Curve type (High or Low), and FE Curve (FE_2 or FE_Total) #csv of samples as list not plate with columns: Date, Plate, Well_ID, SampleID, Dil_Factor, Abs1, Abs2 ; these are Date of run, Plate ID, Well ID, Sample ID, Dilution factor, Absorbance 1 (Fe2 or first read), and Absorbance 2 (feTot or second read) # I save these files in a folder for each day of FE runs

Read in Standards and subset

```
stds <- read.csv("Raw Data/FE_20221220_Stds.csv")
head(stds)
```

```
##   Fe.II.       Abs Curve       FE
## 1    500 1.4255000  High FE_Total
## 2    250 0.9125000  High FE_Total
## 3    100 0.4245000  High FE_Total
## 4     60 0.3273333  High FE_Total
## 5     45 0.2476667  High FE_Total
## 6     30 0.1900000  High FE_Total
```

```
#subset by FE_2 and FE_Total
FE_2_stds <- stds[stds$FE == "FE_2", ]                                    # One factor level
head(FE_2_stds)
```

```
##    Fe.II.    Abs Curve   FE
## 64    500 1.3170  High FE_2
## 65    250 0.7080  High FE_2
## 66    100 0.2730  High FE_2
## 67     60 0.2150  High FE_2
## 68     45 0.1710  High FE_2
## 69     30 0.1205  High FE_2
```

```r
FE_Tot_stds <- stds[stds$FE == "FE_Total", ]                              # One factor level
head(FE_Tot_stds)
```

```
##   Fe.II.       Abs Curve       FE
## 1    500 1.4255000  High FE_Total
## 2    250 0.9125000  High FE_Total
## 3    100 0.4245000  High FE_Total
## 4     60 0.3273333  High FE_Total
## 5     45 0.2476667  High FE_Total
## 6     30 0.1900000  High FE_Total
```

```r
#subset curves for high and low
FE_2_stds_high <-  FE_2_stds[FE_2_stds$Curve == "High", ]                             # One factor le
head(FE_2_stds_high)
```

```
##    Fe.II.    Abs Curve   FE
## 64    500 1.3170  High FE_2
## 65    250 0.7080  High FE_2
## 66    100 0.2730  High FE_2
## 67     60 0.2150  High FE_2
## 68     45 0.1710  High FE_2
## 69     30 0.1205  High FE_2
```

```r
FE_2_stds_low <-  FE_2_stds[FE_2_stds$Curve == "Low", ]                               # One factor leve
head(FE_2_stds_low)
```

```
##     Fe.II.        Abs Curve   FE
## 100     60 0.21500000   Low FE_2
## 101     45 0.17100000   Low FE_2
## 102     30 0.12050000   Low FE_2
## 103     18 0.09300000   Low FE_2
## 104      9 0.06400000   Low FE_2
## 105      6 0.05366667   Low FE_2
```

```r
FE_Tot_stds_high <-  FE_Tot_stds[FE_Tot_stds$Curve == "High", ]                         # One fac
head(FE_Tot_stds_high)
```

```
##   Fe.II.       Abs Curve       FE
## 1    500 1.4255000  High FE_Total
## 2    250 0.9125000  High FE_Total
## 3    100 0.4245000  High FE_Total
## 4     60 0.3273333  High FE_Total
## 5     45 0.2476667  High FE_Total
## 6     30 0.1900000  High FE_Total
```

```r
FE_Tot_stds_low <-  FE_Tot_stds[FE_Tot_stds$Curve == "Low", ]                           # One facto
head(FE_Tot_stds_low)
```

```
##    Fe.II.        Abs Curve       FE
## 37     60 0.32733333   Low FE_Total
```

```
## 38      45 0.24766667    Low FE_Total
## 39      30 0.19000000    Low FE_Total
## 40      18 0.13066667    Low FE_Total
## 41       9 0.07833333    Low FE_Total
## 42       6 0.06466667    Low FE_Total
```

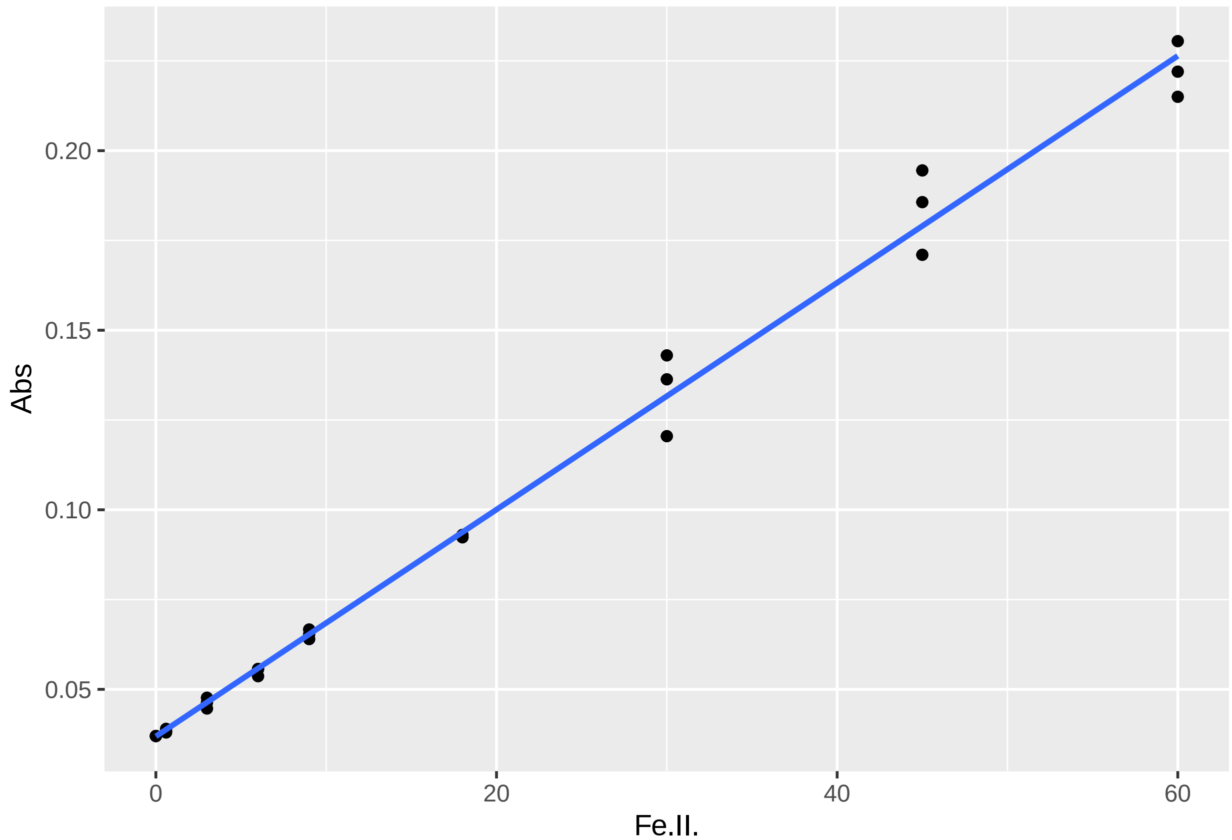
Plot standards

```r
#Plot each of these and calculate the slope, intercept, and R2
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```r
#Low curve for Fe(II)
FE_2_Low <- ggplot(FE_2_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = FA
FE_2_Low
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
FE_2_Low_lm <- lm(FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
summary(FE_2_Low_lm)
```

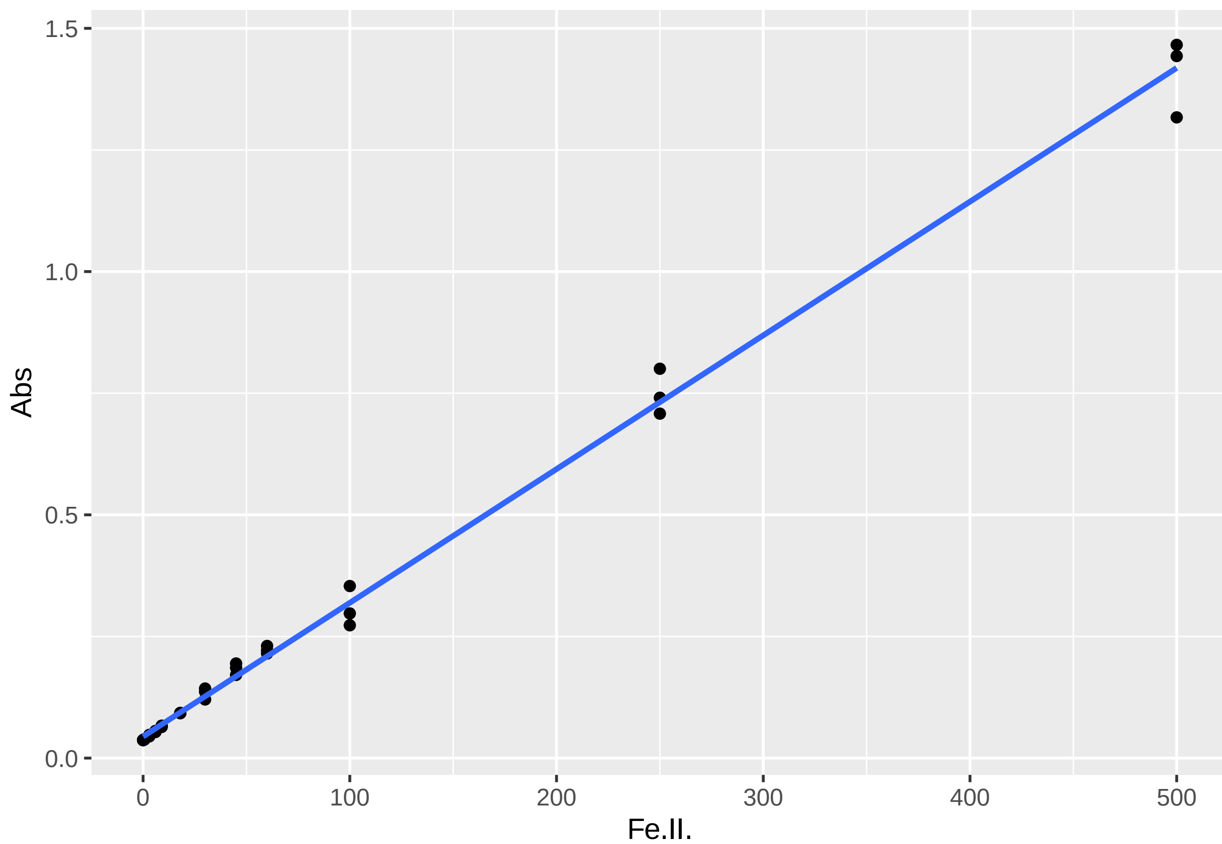
```
##
## Call:
## lm(formula = FE_2_stds_low$Abs ~ FE_2_stds_low$Fe.II.)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0113961 -0.0013766 -0.0001906  0.0007402  0.0154758
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.691e-02  1.501e-03   24.59   <2e-16 ***
## FE_2_stds_low$Fe.II. 3.158e-03  5.391e-05   58.58   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.005682 on 25 degrees of freedom
## Multiple R-squared:  0.9928, Adjusted R-squared:  0.9925
## F-statistic:  3432 on 1 and 25 DF,  p-value: < 2.2e-16
```

```r
cf <- coef(FE_2_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes1 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes1$Curve <- "FE_2_Low"
Slopes1$R2 <- summary(FE_2_Low_lm)$adj.r.squared
Slopes1$Slope <- cf[2]
Slopes1$Intercept <- cf[1]
head(Slopes1)
```

```
##      Curve        R2       Slope  Intercept
## 1 FE_2_Low 0.9924789 0.003158126 0.03690853
```

```r
#High curve for Fe(III)
FE_2_High <- ggplot(FE_2_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se = 
FE_2_High
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
FE_2_High_lm <- lm(FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
summary(FE_2_High_lm)
```

```
##
## Call:
## lm(formula = FE_2_stds_high$Abs ~ FE_2_stds_high$Fe.II.)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.101648 -0.007378 -0.004369  0.010302  0.068820
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           0.0443781  0.0052697   8.421 7.82e-10 ***
## FE_2_stds_high$Fe.II. 0.0027485  0.0000318  86.425  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02713 on 34 degrees of freedom
## Multiple R-squared:  0.9955, Adjusted R-squared:  0.9953
## F-statistic:  7469 on 1 and 34 DF,  p-value: < 2.2e-16
```

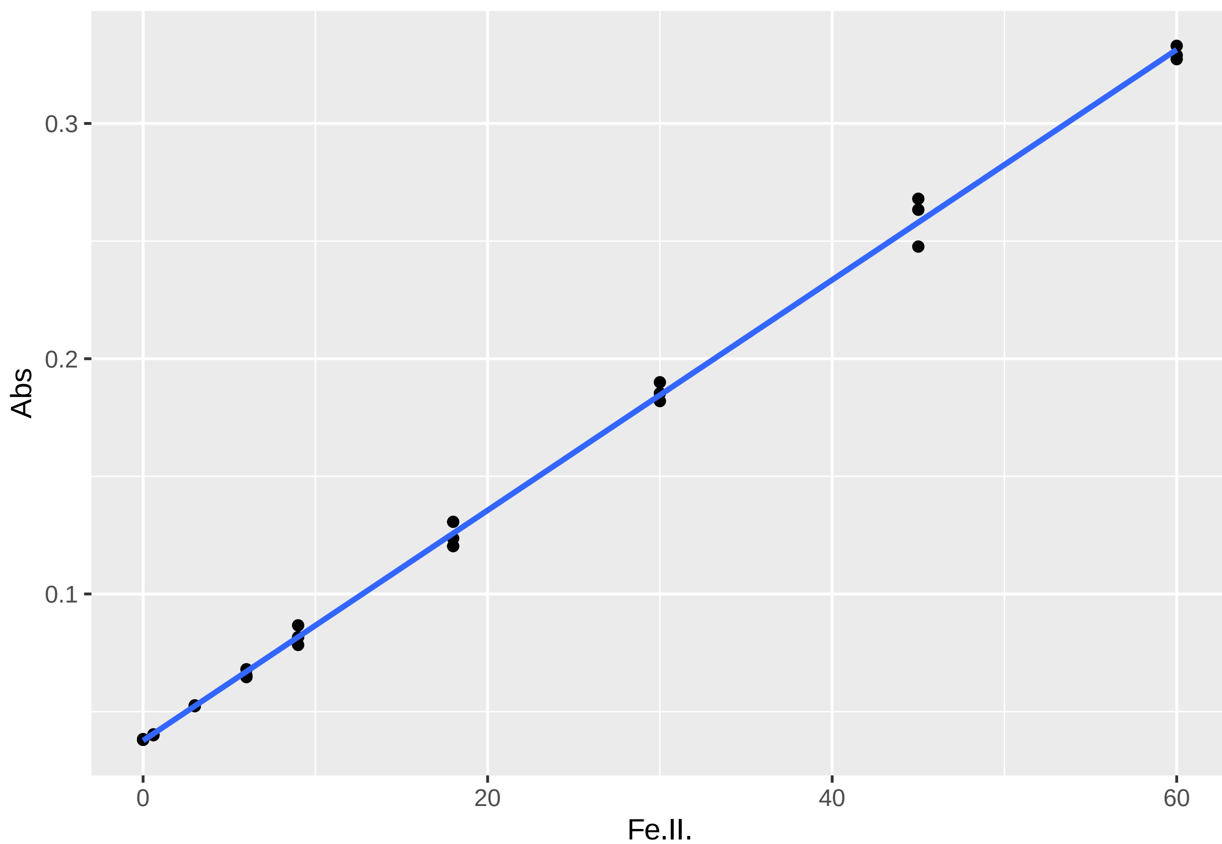
```
cf <- coef(FE_2_High_lm)

#create data frame with 1 rows and 0 columns
```

```r
Slopes2 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes2$Curve <- "FE_2_High"
Slopes2$R2 <- summary(FE_2_High_lm)$adj.r.squared
Slopes2$Slope <- cf[2]
Slopes2$Intercept <- cf[1]
head(Slopes2)
```

```
##       Curve        R2       Slope  Intercept
## 1 FE_2_High 0.9953354 0.002748539 0.04437807
```

```r
#Low curve for Fe(Total)
FE_Tot_Low <- ggplot(FE_Tot_stds_low, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se =
FE_Tot_Low
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
FE_Tot_Low_lm <- lm(FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
summary(FE_Tot_Low_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_low$Abs ~ FE_Tot_stds_low$Fe.II.)
##
```

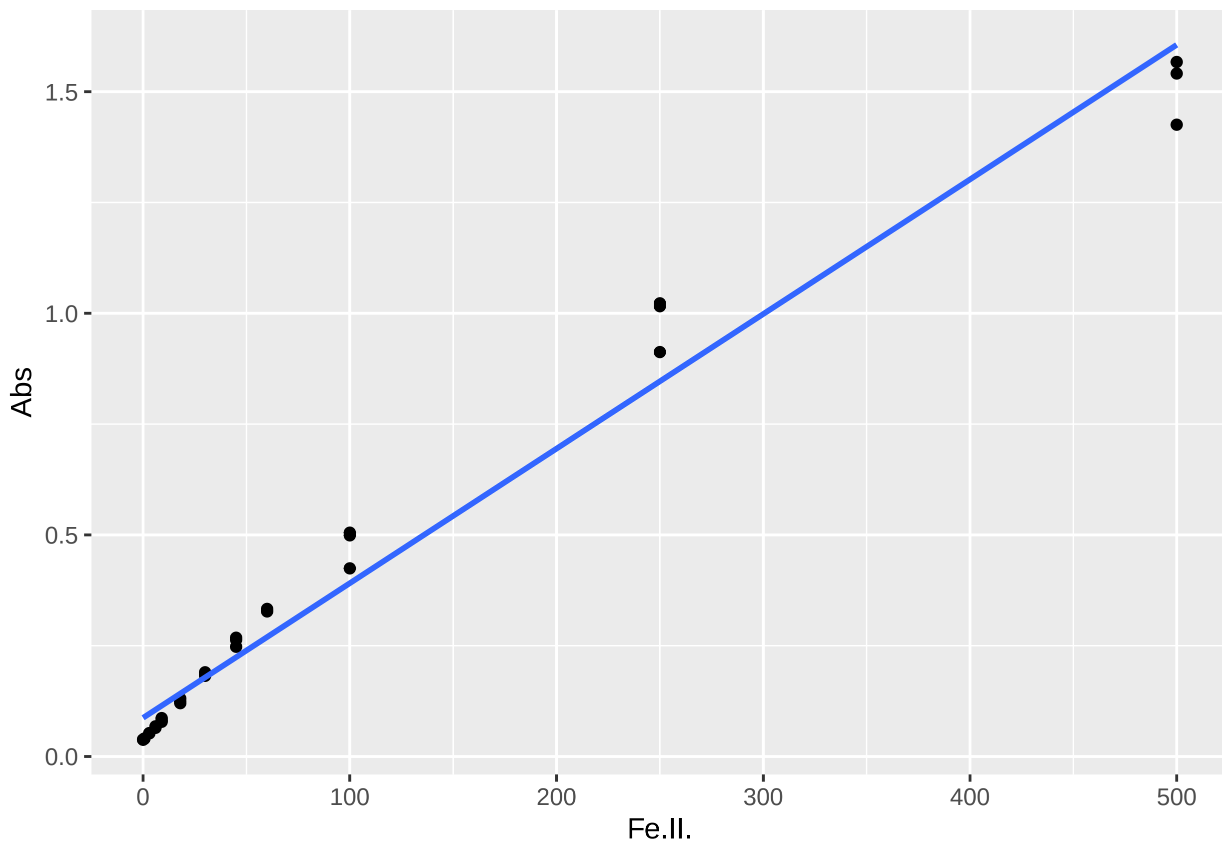
```
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0102915 -0.0022381 -0.0000114  0.0008886  0.0100418
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            3.766e-02  1.060e-03   35.54   <2e-16 ***
## FE_Tot_stds_low$Fe.II. 4.896e-03  3.806e-05  128.63   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.004012 on 25 degrees of freedom
## Multiple R-squared:  0.9985, Adjusted R-squared:  0.9984
## F-statistic: 1.654e+04 on 1 and 25 DF,  p-value: < 2.2e-16
```

```r
cf <- coef(FE_Tot_Low_lm)

#create data frame with 1 rows and 0 columns
Slopes3 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes3$Curve <- "FE_Tot_Low"
Slopes3$R2 <- summary(FE_Tot_Low_lm)$adj.r.squared
Slopes3$Slope <- cf[2]
Slopes3$Intercept <- cf[1]
head(Slopes3)
```

```
##        Curve        R2       Slope  Intercept
## 1 FE_Tot_Low 0.9984309 0.004895559 0.03765801
```

```r
#High curve for Fe(III)
FE_Tot_High <- ggplot(FE_Tot_stds_high, aes(Fe.II., Abs)) + geom_point() + geom_smooth(method = "lm", se
FE_Tot_High
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
FE_Tot_High_lm <- lm(FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
summary(FE_Tot_High_lm)
```

```
##
## Call:
## lm(formula = FE_Tot_stds_high$Abs ~ FE_Tot_stds_high$Fe.II.)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.18018 -0.04408 -0.02480  0.04054  0.17601
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             8.730e-02  1.383e-02   6.314 3.38e-07 ***
## FE_Tot_stds_high$Fe.II. 3.037e-03  8.345e-05  36.391  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07118 on 34 degrees of freedom
## Multiple R-squared:  0.975,  Adjusted R-squared:  0.9742
## F-statistic:  1324 on 1 and 34 DF,  p-value: < 2.2e-16
```

```
cf <- coef(FE_Tot_High_lm)

#create data frame with 1 rows and 0 columns
```

```
Slopes4 <- data.frame(matrix(ncol = 0, nrow = 1))
Slopes4$Curve <- "FE_Tot_High"
Slopes4$R2 <- summary(FE_Tot_High_lm)$adj.r.squared
Slopes4$Slope <- cf[2]
Slopes4$Intercept <- cf[1]
head(Slopes4)
```

```
##         Curve        R2       Slope  Intercept
## 1 FE_Tot_High 0.9742328 0.003036759 0.08730387
```

```
#pull all the slopes together into one dataframe
Slopes <- rbind(Slopes1, Slopes2, Slopes3, Slopes4)


####
```

Read in Sample Data subset

```
dat <- read.csv("Raw Data/FE_20221220_Samples.csv")
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400
## 4 20221220 Dilutions2       4       46          7 1.196 1.303
## 5 20221220 Dilutions2       5       46          7 1.030 1.338
## 6 20221220 Dilutions2       6       46          7 1.008 1.115    BR
```

```
## remove any lines that say stds in SampleID
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
dat %>%  filter(!SampleID =='stds')
```

```
##         Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes
## 1   20221220 Dilutions2       4       46          7 1.196 1.303
## 2   20221220 Dilutions2       5       46          7 1.030 1.338
```

```
## 3   20221220 Dilutions2       6        46          7 1.008 1.115    BR
## 4   20221220 Dilutions2       7        62          7 0.987 1.182
## 5   20221220 Dilutions2       8        62          7 1.047 1.180
## 6   20221220 Dilutions2       9        62          7 0.866 1.107    BR
## 7   20221220 Dilutions2      10        69          7 0.192 0.262
## 8   20221220 Dilutions2      11        69          7 0.176 0.241
## 9   20221220 Dilutions2      12        69          7 0.154 0.227    BR
## 10  20221220 Dilutions2      16        51          7 0.226 0.325
## 11  20221220 Dilutions2      17        51          7 0.217 0.312
## 12  20221220 Dilutions2      18        51          7 0.204 0.315
## 13  20221220 Dilutions2      19        63          7 0.289 0.381
## 14  20221220 Dilutions2      20        63          7 0.275 0.349
## 15  20221220 Dilutions2      21        63          7 0.239 0.347
## 16  20221220 Dilutions2      22        70          7 0.977 1.111
## 17  20221220 Dilutions2      23        70          7 0.902 1.064    BR
## 18  20221220 Dilutions2      24        70          7 1.005 1.273
## 19  20221220 Dilutions2      28        54          7 0.175 0.290    BR
## 20  20221220 Dilutions2      29        54          7 0.150 0.251
## 21  20221220 Dilutions2      30        54          7 0.166 0.247
## 22  20221220 Dilutions2      31        64          7 0.868 1.086
## 23  20221220 Dilutions2      32        64          7 0.800 1.024
## 24  20221220 Dilutions2      33        64          7 0.627 0.969    BR
## 25  20221220 Dilutions2      34        71          7 2.072 1.920    ND
## 26  20221220 Dilutions2      35        71          7 2.021 1.931    ND
## 27  20221220 Dilutions2      36        71          7 1.759 1.804    ND
## 28  20221220 Dilutions2      40        55          7 0.206 0.304    BR
## 29  20221220 Dilutions2      41        55          7 0.187 0.295
## 30  20221220 Dilutions2      42        55          7 0.181 0.278
## 31  20221220 Dilutions2      43       64d          7 0.744 0.933
## 32  20221220 Dilutions2      44       64d          7 0.773 0.949
## 33  20221220 Dilutions2      45       64d          7 0.510 0.688    BR
## 34  20221220 Dilutions2      46        72          7 0.504 0.704
## 35  20221220 Dilutions2      47        72          7 0.481 0.653
## 36  20221220 Dilutions2      48        72          7 0.469 0.569    BR
## 37  20221220 Dilutions2      52        58          7 1.283 1.399    BR
## 38  20221220 Dilutions2      53        58          7 1.013 1.163
## 39  20221220 Dilutions2      54        58          7 1.081 1.221
## 40  20221220 Dilutions2      55        65          7 1.920 1.894    ND
## 41  20221220 Dilutions2      56        65          7 1.607 1.561    ND
## 42  20221220 Dilutions2      57        65          7 1.789 1.806    ND
## 43  20221220 Dilutions2      58        73          7 0.936 1.114    BR
## 44  20221220 Dilutions2      59        73          7 0.691 0.875
## 45  20221220 Dilutions2      60        73          7 0.741 0.974
## 46  20221220 Dilutions2      64        59          7 0.453 0.605
## 47  20221220 Dilutions2      65        59          7 0.433 0.606
## 48  20221220 Dilutions2      66        59          7 0.401 0.533    BR
## 49  20221220 Dilutions2      68     65spk          7 1.570 1.540    ND
## 50  20221220 Dilutions2      69     65spk          7 1.584 1.556    ND
## 51  20221220 Dilutions2      70     65spk          7 1.667 1.630    ND
## 52  20221220 Dilutions2      71        74          7 1.620 1.637    ND
## 53  20221220 Dilutions2      72        74          7 1.651 1.604    ND
## 54  20221220 Dilutions2      76        60          7 0.187 0.274
## 55  20221220 Dilutions2      77        60          7 0.184 0.273
## 56  20221220 Dilutions2      78        60          7 0.162 0.253    BR
```

```
## 57  20221220 Dilutions2       80        67           7 1.346 1.380    ND
## 58  20221220 Dilutions2       81        67           7 1.306 1.494    ND
## 59  20221220 Dilutions2       82        67           7 1.407 1.545    ND
## 60  20221220 Dilutions2       83        82           7 0.310 0.471    BR
## 61  20221220 Dilutions2       84        82           7 0.287 0.411
## 62  20221220 Dilutions2       89        61           7 0.429 0.645
## 63  20221220 Dilutions2       90        61           7 0.466 0.635
## 64  20221220 Dilutions2       91        61           7 0.347 0.599    BR
## 65  20221220 Dilutions2       92        68           7 1.534 1.526    ND
## 66  20221220 Dilutions2       93        68           7 1.180 1.164    ND
## 67  20221220 Dilutions2       94        68           7 1.211 1.327    ND
## 68  20221220 Dilutions2       95        74           7 1.291 1.413    ND
## 69  20221220 Dilutions2       96        82           7 0.277 0.398
## 70  20221220 Dilutions3        4         5           7 0.276 0.433
## 71  20221220 Dilutions3        5         5           7 0.261 0.390
## 72  20221220 Dilutions3        6         5           7 0.216 0.366    BR
## 73  20221220 Dilutions3        7        29           7 0.455 1.012    BR
## 74  20221220 Dilutions3        8        29           7 0.404 0.957
## 75  20221220 Dilutions3        9        29           7 0.418 0.945
## 76  20221220 Dilutions3       10        38           7 1.943 0.467
## 77  20221220 Dilutions3       11        38           7 1.431 0.466    BR
## 78  20221220 Dilutions3       12        38           7 1.723 0.411
## 79  20221220 Dilutions3       16         8           7 0.302 0.698
## 80  20221220 Dilutions3       17         8           7 0.304 0.610
## 81  20221220 Dilutions3       18         8           7 0.271 0.615
## 82  20221220 Dilutions3       19        30           7 0.822 0.727
## 83  20221220 Dilutions3       20        30           7 0.866 0.777    BR
## 84  20221220 Dilutions3       21        30           7 0.716 0.701
## 85  20221220 Dilutions3       22        39           7 0.311 0.553    BR
## 86  20221220 Dilutions3       23        39           7 0.294 0.475
## 87  20221220 Dilutions3       24        39           7 0.301 0.444
## 88  20221220 Dilutions3       28         9           7 0.525 0.084    BR
## 89  20221220 Dilutions3       29         9           7 0.430 0.082
## 90  20221220 Dilutions3       30         9           7 0.462 0.080
## 91  20221220 Dilutions3       31        31           7 0.540 0.866
## 92  20221220 Dilutions3       32        31           7 0.600 0.739    BR
## 93  20221220 Dilutions3       33        31           7 0.519 0.851
## 94  20221220 Dilutions3       34        40           7 0.401 1.271    BR
## 95  20221220 Dilutions3       35        40           7 0.329 1.345
## 96  20221220 Dilutions3       36        40           7 0.320 1.051
## 97  20221220 Dilutions3       40        10           7 0.069 0.262
## 98  20221220 Dilutions3       41        10           7 0.065 0.291
## 99  20221220 Dilutions3       42        10           7 0.063 0.279
## 100 20221220 Dilutions3       43       31d           7 0.649 1.051    BR
## 101 20221220 Dilutions3       44       31d           7 0.577 0.985
## 102 20221220 Dilutions3       45       31d           7 0.559 0.846
## 103 20221220 Dilutions3       46        41           7 1.112 0.243
## 104 20221220 Dilutions3       47        41           7 1.212 0.248
## 105 20221220 Dilutions3       48        41           7 0.974 0.243    BR
## 106 20221220 Dilutions3       52        11           7 0.204 0.399    BR
## 107 20221220 Dilutions3       53        11           7 0.179 0.408
## 108 20221220 Dilutions3       54        11           7 0.172 0.421
## 109 20221220 Dilutions3       55        32           7 0.835 0.757
## 110 20221220 Dilutions3       56        32           7 0.782 0.814
```

```
## 111 20221220 Dilutions3      57        32           7 0.624 0.697    BR
## 112 20221220 Dilutions3      58        42           7 0.180 0.528    BR
## 113 20221220 Dilutions3      59        42           7 0.157 0.483
## 114 20221220 Dilutions3      60        42           7 0.165 0.391
## 115 20221220 Dilutions3      64        12           7 0.309 0.675    BR
## 116 20221220 Dilutions3      65        12           7 0.270 0.585
## 117 20221220 Dilutions3      66        12           7 0.279 0.567
## 118 20221220 Dilutions3      67     32spk           7 0.621 1.203
## 119 20221220 Dilutions3      68     32spk           7 0.724 1.287    BR
## 120 20221220 Dilutions3      69     32spk           7 0.615 1.167
## 121 20221220 Dilutions3      70        43           7 0.352 1.016
## 122 20221220 Dilutions3      71        43           7 0.279 1.031    BR
## 123 20221220 Dilutions3      72        43           7 0.324 1.063
## 124 20221220 Dilutions3      76        26           7 0.549 0.928    BR
## 125 20221220 Dilutions3      77        26           7 0.427 0.887
## 126 20221220 Dilutions3      78        26           7 0.415 0.816
## 127 20221220 Dilutions3      79        35           7 1.147 0.399
## 128 20221220 Dilutions3      80        35           7 1.220 0.365
## 129 20221220 Dilutions3      81        35           7 1.022 0.308    BR
## 130 20221220 Dilutions3      82        44           7 0.898 0.836
## 131 20221220 Dilutions3      83        44           7 0.840 0.601    BR
## 132 20221220 Dilutions3      84        44           7 0.945 0.733
## 133 20221220 Dilutions3      88        28           7 0.698 0.000
## 134 20221220 Dilutions3      89        28           7 0.630 0.000
## 135 20221220 Dilutions3      90        28           7 0.561 0.000
## 136 20221220 Dilutions3      91        37           7 0.257 0.000
## 137 20221220 Dilutions3      92        37           7 0.264 0.000
## 138 20221220 Dilutions3      93        37           7 0.182 0.000    BR
## 139 20221220 Dilutions3      94        45           7 0.481 0.000
## 140 20221220 Dilutions3      95        45           7 0.426 0.000
## 141 20221220 Dilutions3      96        45           7 0.549 0.000
## 142 20221220   Samples1       4        89           7 0.094 0.137
## 143 20221220   Samples1       5        89           7 0.086 0.120
## 144 20221220   Samples1       6        89           7 0.091 0.119
## 145 20221220   Samples1       7        97           7 0.043 0.049
## 146 20221220   Samples1       8        97           7 0.042 0.048
## 147 20221220   Samples1       9        97           7 0.042 0.048
## 148 20221220   Samples1      10       103           1 0.069 0.070
## 149 20221220   Samples1      11       103           1 0.066 0.068
## 150 20221220   Samples1      12       103           1 0.065 0.067
## 151 20221220   Samples1      16        90           7 0.050 0.057
## 152 20221220   Samples1      17        90           7 0.050 0.055
## 153 20221220   Samples1      18        90           7 0.048 0.054
## 154 20221220   Samples1      19        98           7 0.061 0.081
## 155 20221220   Samples1      20        98           7 0.065 0.078
## 156 20221220   Samples1      21        98           7 0.063 0.085
## 157 20221220   Samples1      22       104           1 0.082 0.081
## 158 20221220   Samples1      23       104           1 0.079 0.078
## 159 20221220   Samples1      24       104           1 0.080 0.078
## 160 20221220   Samples1      28        91           7 0.052 0.067
## 161 20221220   Samples1      29        91           7 0.055 0.066
## 162 20221220   Samples1      30        91           7 0.055 0.065
## 163 20221220   Samples1      31        99           7 0.094 0.132
## 164 20221220   Samples1      32        99           7 0.088 0.119
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## 165 20221220   Samples1      33        99          7 0.087 0.140
## 166 20221220   Samples1      34       105          1 0.040 0.048
## 167 20221220   Samples1      35       105          1 0.040 0.047
## 168 20221220   Samples1      36       105          1 0.040 0.046
## 169 20221220   Samples1      40        92          7 0.173 0.288
## 170 20221220   Samples1      41        92          7 0.166 0.239
## 171 20221220   Samples1      42        92          7 0.170 0.223
## 172 20221220   Samples1      43       99d          7 0.086 0.121
## 173 20221220   Samples1      44       99d          7 0.086 0.113
## 174 20221220   Samples1      45       99d          7 0.092 0.142
## 175 20221220   Samples1      46       106          1 0.078 0.079
## 176 20221220   Samples1      47       106          1 0.075 0.076
## 177 20221220   Samples1      48       106          1 0.073 0.074
## 178 20221220   Samples1      52        93          7 0.045 0.055
## 179 20221220   Samples1      53        93          7 0.047 0.053
## 180 20221220   Samples1      54        93          7 0.046 0.051
## 181 20221220   Samples1      55       100          1 0.085 0.085
## 182 20221220   Samples1      56       100          1 0.082 0.083
## 183 20221220   Samples1      57       100          1 0.081 0.083
## 184 20221220   Samples1      58       107          1 0.073 0.074
## 185 20221220   Samples1      59       107          1 0.071 0.073
## 186 20221220   Samples1      60       107          1 0.070 0.070
## 187 20221220   Samples1      64        94          7 0.061 0.080
## 188 20221220   Samples1      65        94          7 0.061 0.078
## 189 20221220   Samples1      66        94          7 0.060 0.076
## 190 20221220   Samples1      67    100spk          1 0.164 0.277
## 191 20221220   Samples1      68    100spk          1 0.169 0.251
## 192 20221220   Samples1      69    100spk          1 0.159 0.216     BR
## 193 20221220   Samples1      70       108          7 0.054 0.087
## 194 20221220   Samples1      71       108          7 0.055 0.087
## 195 20221220   Samples1      72       108          7 0.054 0.071
## 196 20221220   Samples1      76        95          7 0.107 0.189     BR
## 197 20221220   Samples1      77        95          7 0.087 0.120
## 198 20221220   Samples1      78        95          7 0.102 0.129
## 199 20221220   Samples1      79       101          1 0.064 0.065
## 200 20221220   Samples1      80       101          1 0.063 0.065
## 201 20221220   Samples1      81       101          1 0.062 0.064
## 202 20221220   Samples1      82       109          7 0.044 0.089
## 203 20221220   Samples1      83       109          7 0.043 0.100
## 204 20221220   Samples1      84       109          7 0.044 0.085
## 205 20221220   Samples1      88        96          7 0.061 0.095
## 206 20221220   Samples1      89        96          7 0.060 0.085
## 207 20221220   Samples1      90        96          7 0.063 0.087
## 208 20221220   Samples1      91       102          1 0.068 0.070
## 209 20221220   Samples1      92       102          1 0.067 0.066
## 210 20221220   Samples1      93       102          1 0.067 0.068
## 211 20221220   Samples1      94       110          7 0.056 0.074     BR
## 212 20221220   Samples1      95       110          7 0.055 0.256
## 213 20221220   Samples1      96       110          7 0.057 0.216
## 214 20221220   Samples2       4       111          7 0.060 0.110
## 215 20221220   Samples2       5       111          7 0.058 0.105
## 216 20221220   Samples2       6       111          7 0.056 0.099
## 217 20221220   Samples2       7       119          7 0.499 0.694
## 218 20221220   Samples2       8       119          7 0.527 0.704
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## 219 20221220   Samples2       9       119          7 0.534 0.756    BR
## 220 20221220   Samples2      10       125          7 1.190 1.271    BR
## 221 20221220   Samples2      11       125          7 0.978 1.082
## 222 20221220   Samples2      12       125          7 0.788 0.899
## 223 20221220   Samples2      16       112          7 1.451 1.532
## 224 20221220   Samples2      17       112          7 1.566 1.624    BR
## 225 20221220   Samples2      18       112          7 1.364 1.535
## 226 20221220   Samples2      19       120          7 0.146 0.233
## 227 20221220   Samples2      20       120          7 0.132 0.201
## 228 20221220   Samples2      21       120          7 0.133 0.213
## 229 20221220   Samples2      22       126          7 1.080 1.234
## 230 20221220   Samples2      23       126          7 1.075 1.117
## 231 20221220   Samples2      24       126          7 1.051 1.175
## 232 20221220   Samples2      28       113          7 1.387 1.449    BR
## 233 20221220   Samples2      29       113          7 1.159 1.229
## 234 20221220   Samples2      30       113          7 1.197 1.308
## 235 20221220   Samples2      31       121          7 0.364 0.505
## 236 20221220   Samples2      32       121          7 0.325 0.456
## 237 20221220   Samples2      33       121          7 0.221 0.363    BR
## 238 20221220   Samples2      34       127          7 1.138 1.196
## 239 20221220   Samples2      35       127          7 1.065 1.124
## 240 20221220   Samples2      36       127          7 0.926 0.999    BR
## 241 20221220   Samples2      40       114          7 0.927 1.077
## 242 20221220   Samples2      41       114          7 1.005 1.111
## 243 20221220   Samples2      42       114          7 0.861 1.083    BR
## 244 20221220   Samples2      43      121d          7 0.322 0.472
## 245 20221220   Samples2      44      121d          7 0.334 0.503
## 246 20221220   Samples2      45      121d          7 0.228 0.402    BR
## 247 20221220   Samples2      46       128          7 1.103 1.269
## 248 20221220   Samples2      47       128          7 1.246 1.299
## 249 20221220   Samples2      48       128          7 0.823 0.936    BR
## 250 20221220   Samples2      52       115          7 0.620 0.757
## 251 20221220   Samples2      53       115          7 0.604 0.783
## 252 20221220   Samples2      54       115          7 0.555 0.763    BR
## 253 20221220   Samples2      55       122          7 0.811 0.937
## 254 20221220   Samples2      56       122          7 0.768 0.885    BR
## 255 20221220   Samples2      57       122          7 0.810 0.932
## 256 20221220   Samples2      58       129          7 1.053 1.131
## 257 20221220   Samples2      59       129          7 1.009 1.032
## 258 20221220   Samples2      60       129          7 0.805 0.930    BR
## 259 20221220   Samples2      64       116          7 0.538 0.680
## 260 20221220   Samples2      65       116          7 0.543 0.708
## 261 20221220   Samples2      66       116          7 0.514 0.683
## 262 20221220   Samples2      67    122spk          7 0.701 0.794
## 263 20221220   Samples2      68    122spk          7 0.747 0.810
## 264 20221220   Samples2      69    122spk          7 0.676 0.839
## 265 20221220   Samples2      70       130          7 1.154 1.250
## 266 20221220   Samples2      71       130          7 1.017 1.104
## 267 20221220   Samples2      72       130          7 0.998 1.087
## 268 20221220   Samples2      76       117          7 0.573 0.781
## 269 20221220   Samples2      77       117          7 0.410 0.589
## 270 20221220   Samples2      78       117          7 0.470 0.658
## 271 20221220   Samples2      79       123          7 1.225 1.280
## 272 20221220   Samples2      80       123          7 0.923 1.016
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## 273 20221220   Samples2      81       123           7 0.742 0.879     BR
## 274 20221220   Samples2      82       131           7 0.624 0.794
## 275 20221220   Samples2      83       131           7 0.528 0.656     BR
## 276 20221220   Samples2      84       131           7 0.627 0.769
## 277 20221220   Samples2      88       118           7 0.501 0.714
## 278 20221220   Samples2      89       118           7 0.530 0.755
## 279 20221220   Samples2      90       118           7 0.445 0.630
## 280 20221220   Samples2      91       124           7 1.159 1.246
## 281 20221220   Samples2      92       124           7 0.880 0.983     BR
## 282 20221220   Samples2      93       124           7 0.962 1.111
## 283 20221220   Samples2      94       132           1 0.179 0.316
## 284 20221220   Samples2      95       132           1 0.169 0.247     BR
## 285 20221220   Samples2      96       132           1 0.187 0.325
## 286 20221220   Samples3       4       133           1 0.126 0.355     BR
## 287 20221220   Samples3       5       133           1 0.149 0.444
## 288 20221220   Samples3       6       133           1 0.151 0.558
## 289 20221220   Samples3       7       141           1 1.246 1.548
## 290 20221220   Samples3       8       141           1 1.136 1.779     BR
## 291 20221220   Samples3       9       141           1 1.266 1.537
## 292 20221220   Samples3      10       147           7 0.190 0.286
## 293 20221220   Samples3      11       147           7 0.146 0.231     BR
## 294 20221220   Samples3      12       147           7 0.213 0.288
## 295 20221220   Samples3      16       134           1 0.336 0.454
## 296 20221220   Samples3      17       134           1 0.337 0.488
## 297 20221220   Samples3      18       134           1 0.307 0.455
## 298 20221220   Samples3      19       142           1 1.345 1.484
## 299 20221220   Samples3      20       142           1 1.341 1.531
## 300 20221220   Samples3      21       142           1 1.451 1.619     BR
## 301 20221220   Samples3      22       148           7 0.049 0.058
## 302 20221220   Samples3      23       148           7 0.049 0.055
## 303 20221220   Samples3      24       148           7 0.048 0.056
## 304 20221220   Samples3      28       135           1 1.080 1.299
## 305 20221220   Samples3      29       135           1 0.816 1.456     BR
## 306 20221220   Samples3      30       135           1 1.033 1.485
## 307 20221220   Samples3      31       143           1 1.510 1.710     BR
## 308 20221220   Samples3      32       143           1 1.062 1.401
## 309 20221220   Samples3      33       143           1 1.265 1.363
## 310 20221220   Samples3      34       149           7 0.255 0.365     BR
## 311 20221220   Samples3      35       149           7 0.180 0.254
## 312 20221220   Samples3      36       149           7 0.196 0.278
## 313 20221220   Samples3      40       136           1 0.820 1.182
## 314 20221220   Samples3      41       136           1 0.786 1.046
## 315 20221220   Samples3      42       136           1 0.935 1.168     BR
## 316 20221220   Samples3      43      143d           1 1.226 1.497
## 317 20221220   Samples3      44      143d           1 1.305 1.572     BR
## 318 20221220   Samples3      45      143d           1 1.231 1.499     BR
## 319 20221220   Samples3      46       150           7 0.652 0.819
## 320 20221220   Samples3      47       150           7 0.619 0.789
## 321 20221220   Samples3      48       150           7 0.585 0.720
## 322 20221220   Samples3      52       137           1 0.826 1.171
## 323 20221220   Samples3      53       137           1 1.075 1.250     BR
## 324 20221220   Samples3      54       137           1 0.943 1.103
## 325 20221220   Samples3      55       144           1 0.072 0.089
## 326 20221220   Samples3      56       144           1 0.075 0.089
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## 327 20221220   Samples3      57      144          1 0.078 0.087
## 328 20221220   Samples3      58      151          7 0.049 0.054
## 329 20221220   Samples3      59      151          7 0.048 0.053
## 330 20221220   Samples3      60      151          7 0.048 0.053
## 331 20221220   Samples3      64      138          1 1.298 1.580
## 332 20221220   Samples3      65      138          1 1.095 1.496    BR
## 333 20221220   Samples3      66      138          1 1.226 1.582
## 334 20221220   Samples3      67   144spk          1 0.061 0.080
## 335 20221220   Samples3      68   144spk          1 0.061 0.085
## 336 20221220   Samples3      69   144spk          1 0.059 0.084
## 337 20221220   Samples3      70      152          7 0.177 0.277
## 338 20221220   Samples3      71      152          7 0.159 0.242    BR
## 339 20221220   Samples3      72      152          7 0.181 0.268
## 340 20221220   Samples3      76      139          1 1.417 1.783
## 341 20221220   Samples3      77      139          1 1.389 1.840
## 342 20221220   Samples3      78      139          1 1.576 1.723    BR
## 343 20221220   Samples3      79      145          1 0.041 0.045
## 344 20221220   Samples3      80      145          1 0.043 0.045
## 345 20221220   Samples3      81      145          1 0.042 0.044
## 346 20221220   Samples3      82      153          7 0.301 0.418
## 347 20221220   Samples3      83      153          7 0.263 0.435
## 348 20221220   Samples3      84      153          7 0.299 0.437
## 349 20221220   Samples3      88      140          1 1.569 1.712
## 350 20221220   Samples3      89      140          1 1.200 1.489    BR
## 351 20221220   Samples3      90      140          1 1.591 1.715
## 352 20221220   Samples3      91      146          7 0.249 0.332    BR
## 353 20221220   Samples3      92      146          7 0.194 0.290
## 354 20221220   Samples3      93      146          7 0.173 0.284
## 355 20221220   Samples3      94      154          7 0.043 0.045
## 356 20221220   Samples3      95      154          7 0.042 0.046
## 357 20221220   Samples3      96      154          7 0.042 0.044
## 358 20221220   Samples4       1     Stds          1 1.308 1.351
## 359 20221220   Samples4       2     Stds          1 1.211 1.335
## 360 20221220   Samples4       3     Stds          1 0.805 1.060
## 361 20221220   Samples4       4      155          1 0.045 0.047
## 362 20221220   Samples4       5      155          1 0.045 0.046
## 363 20221220   Samples4       6      155          1 0.046 0.048
## 364 20221220   Samples4       7      163          1 0.116 0.117
## 365 20221220   Samples4       8      163          1 0.113 0.113
## 366 20221220   Samples4       9      163          1 0.109 0.107    BR
## 367 20221220   Samples4      10      169          7 0.484 0.661
## 368 20221220   Samples4      11      169          7 0.460 0.674
## 369 20221220   Samples4      12      169          7 0.335 0.491    BR
## 370 20221220   Samples4      13     Stds          1 0.063 0.078
## 371 20221220   Samples4      14     Stds          1 0.067 0.080
## 372 20221220   Samples4      15     Stds          1 0.064 0.077
## 373 20221220   Samples4      16      156          1 0.045 0.049
## 374 20221220   Samples4      17      156          1 0.046 0.048
## 375 20221220   Samples4      18      156          1 0.046 0.048
## 376 20221220   Samples4      19      164          1 0.095 0.123    BR
## 377 20221220   Samples4      20      164          1 0.116 0.117
## 378 20221220   Samples4      21      164          1 0.113 0.116
## 379 20221220   Samples4      22      170          7 0.845 1.046
## 380 20221220   Samples4      23      170          7 0.865 1.148
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## 381 20221220   Samples4     24      170          7 0.625 0.897    BR
## 382 20221220   Samples4     25     Stds          1 0.038 0.040
## 383 20221220   Samples4     26     Stds          1 0.039 0.041
## 384 20221220   Samples4     27     Stds          1 0.038 0.040
## 385 20221220   Samples4     28      157          1 0.041 0.044
## 386 20221220   Samples4     29      157          1 0.042 0.043
## 387 20221220   Samples4     30      157          1 0.043 0.044
## 388 20221220   Samples4     31      165          1 0.100 0.108
## 389 20221220   Samples4     32      165          1 0.098 0.103
## 390 20221220   Samples4     33      165          1 0.103 0.102
## 391 20221220   Samples4     34      171          7 0.843 0.923
## 392 20221220   Samples4     35      171          7 0.772 0.940
## 393 20221220   Samples4     36      171          7 0.810 0.974
## 394 20221220   Samples4     37     Stds          1 0.037 0.038
## 395 20221220   Samples4     38     Stds          1 0.037 0.038
## 396 20221220   Samples4     39     Stds          1 0.037 0.038
## 397 20221220   Samples4     40      158          1 0.050 0.052
## 398 20221220   Samples4     41      158          1 0.049 0.051
## 399 20221220   Samples4     42      158          1 0.050 0.050
## 400 20221220   Samples4     43     165d          1 0.108 0.106
## 401 20221220   Samples4     44     165d          1 0.108 0.105
## 402 20221220   Samples4     45     165d          1 0.104 0.103
## 403 20221220   Samples4     46      172          7 0.456 0.575    BR
## 404 20221220   Samples4     47      172          7 0.617 0.751
## 405 20221220   Samples4     48      172          7 0.600 0.757
## 406 20221220   Samples4     49     Stds          1 0.042 0.175
## 407 20221220   Samples4     50     Stds          1 0.041 0.157
## 408 20221220   Samples4     51     Stds          1 0.042 0.165
## 409 20221220   Samples4     52      159          1 0.058 0.063
## 410 20221220   Samples4     53      159          1 0.060 0.065
## 411 20221220   Samples4     54      159          1 0.065 0.067
## 412 20221220   Samples4     55      166          7 0.319 0.430    BR
## 413 20221220   Samples4     56      166          7 0.236 0.350
## 414 20221220   Samples4     57      166          7 0.245 0.356
## 415 20221220   Samples4     58      173          7 0.468 0.643    BR
## 416 20221220   Samples4     59      173          7 0.419 0.572
## 417 20221220   Samples4     60      173          7 0.425 0.598
## 418 20221220   Samples4     61     Stds          1 0.039 0.059
## 419 20221220   Samples4     62     Stds          1 0.039 0.058
## 420 20221220   Samples4     63     Stds          1 0.039 0.056
## 421 20221220   Samples4     64      160          1 0.051 0.052
## 422 20221220   Samples4     65      160          1 0.050 0.051
## 423 20221220   Samples4     66      160          1 0.050 0.051
## 424 20221220   Samples4     67   166spk          7 0.295 0.411
## 425 20221220   Samples4     68   166spk          7 0.294 0.394
## 426 20221220   Samples4     69   166spk          7 0.252 0.398
## 427 20221220   Samples4     70      174          7 0.077 0.099
## 428 20221220   Samples4     71      174          7 0.085 0.103
## 429 20221220   Samples4     72      174          7 0.071 0.085
## 430 20221220   Samples4     73     Stds          1 0.037 0.039
## 431 20221220   Samples4     74     Stds          1 0.040 0.041
## 432 20221220   Samples4     75     Stds          1 0.037 0.039
## 433 20221220   Samples4     76      161          1 0.048 0.049
## 434 20221220   Samples4     77      161          1 0.050 0.050
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## 435 20221220   Samples4      78      161          1 0.050 0.050
## 436 20221220   Samples4      79      167          7 0.310 0.413    BR
## 437 20221220   Samples4      80      167          7 0.223 0.302
## 438 20221220   Samples4      81      167          7 0.231 0.310
## 439 20221220   Samples4      82      175          7 0.042 0.044
## 440 20221220   Samples4      83      175          7 0.042 0.044
## 441 20221220   Samples4      84      175          7 0.042 0.044
## 442 20221220   Samples4      85     Stds          1 0.037 0.038
## 443 20221220   Samples4      86     Stds          1 0.038 0.038
## 444 20221220   Samples4      87     Stds          1 0.039 0.039
## 445 20221220   Samples4      88      162          1 0.042 0.045
## 446 20221220   Samples4      89      162          1 0.042 0.044
## 447 20221220   Samples4      90      162          1 0.042 0.044
## 448 20221220   Samples4      91      168          7 0.157 0.242
## 449 20221220   Samples4      92      168          7 0.156 0.238
## 450 20221220   Samples4      93      168          7 0.131 0.209
## 451 20221220   Samples4      94      176          7 0.916 1.095
## 452 20221220   Samples4      95      176          7 0.811 1.151
## 453 20221220   Samples4      96      176          7 0.815 1.010
## 454 20221220   Samples5       1     Stds          1 1.317 1.349
## 455 20221220   Samples5       2     Stds          1 1.019 1.387    BR
## 456 20221220   Samples5       3     Stds          1 1.320 1.375
## 457 20221220   Samples5       4      177          1 0.053 0.069    ND
## 458 20221220   Samples5       5      177          1 0.050 0.062    ND
## 459 20221220   Samples5       6      177          1 0.055 0.067    ND
## 460 20221220   Samples5       7      185          1 0.345 0.480    BR
## 461 20221220   Samples5       8      185          1 0.385 0.546
## 462 20221220   Samples5       9      185          1 0.375 0.546
## 463 20221220   Samples5      10      191          7 0.058 0.075
## 464 20221220   Samples5      11      191          7 0.054 0.065
## 465 20221220   Samples5      12      191          7 0.055 0.064
## 466 20221220   Samples5      13     Stds          1 0.064 0.083
## 467 20221220   Samples5      14     Stds          1 0.066 0.085
## 468 20221220   Samples5      15     Stds          1 0.061 0.079
## 469 20221220   Samples5      16      178          1 0.196 0.361
## 470 20221220   Samples5      17      178          1 0.249 0.375    BR
## 471 20221220   Samples5      18      178          1 0.184 0.282
## 472 20221220   Samples5      19      186          1 1.091 1.246
## 473 20221220   Samples5      20      186          1 1.144 1.339
## 474 20221220   Samples5      21      186          1 1.035 1.269
## 475 20221220   Samples5      22      192          7 0.271 0.412
## 476 20221220   Samples5      23      192          7 0.283 0.400
## 477 20221220   Samples5      24      192          7 0.228 0.349    BR
## 478 20221220   Samples5      25     Stds          1 0.039 0.041
## 479 20221220   Samples5      26     Stds          1 0.039 0.041
## 480 20221220   Samples5      27     Stds          1 0.039 0.041
## 481 20221220   Samples5      28      179          1 0.428 0.639
## 482 20221220   Samples5      29      179          1 0.418 0.621
## 483 20221220   Samples5      30      179          1 0.408 0.580    BR
## 484 20221220   Samples5      31      187          1 1.093 1.274
## 485 20221220   Samples5      32      187          1 0.978 1.197
## 486 20221220   Samples5      33      187          1 0.713 0.906    BR
## 487 20221220   Samples5      34      193          7 0.061 0.074
## 488 20221220   Samples5      35      193          7 0.058 0.070
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## 489 20221220   Samples5      36       193          7 0.061 0.071
## 490 20221220   Samples5      37      Stds          1 0.037 0.038
## 491 20221220   Samples5      38      Stds          1 0.037 0.038
## 492 20221220   Samples5      39      Stds          1 0.037 0.038
## 493 20221220   Samples5      40       180          1 0.412 0.571     BR
## 494 20221220   Samples5      41       180          1 0.301 0.483
## 495 20221220   Samples5      42       180          1 0.279 0.441
## 496 20221220   Samples5      43      187d          1 1.074 1.339
## 497 20221220   Samples5      44      187d          1 1.072 1.294
## 498 20221220   Samples5      45      187d          1 1.112 1.295
## 499 20221220   Samples5      46       194          7 0.046 0.057
## 500 20221220   Samples5      47       194          7 0.049 0.057
## 501 20221220   Samples5      48       194          7 0.047 0.055
## 502 20221220   Samples5      49      Stds          1 0.041 0.208
## 503 20221220   Samples5      50      Stds          1 0.042 0.211
## 504 20221220   Samples5      51      Stds          1 0.042 0.205
## 505 20221220   Samples5      52       181          1 0.897 1.001
## 506 20221220   Samples5      53       181          1 0.868 1.169
## 507 20221220   Samples5      54       181          1 0.678 0.872     BR
## 508 20221220   Samples5      55       188          7 0.038 0.039
## 509 20221220   Samples5      56       188          7 0.039 0.041
## 510 20221220   Samples5      57       188          7 0.038 0.039
## 511 20221220   Samples5      58       195          7 0.125 0.192
## 512 20221220   Samples5      59       195          7 0.129 0.185
## 513 20221220   Samples5      60       195          7 0.117 0.172     BR
## 514 20221220   Samples5      61      Stds          1 0.039 0.065
## 515 20221220   Samples5      62      Stds          1 0.038 0.063
## 516 20221220   Samples5      63      Stds          1 0.038 0.062
## 517 20221220   Samples5      64       182          1 0.205 0.327
## 518 20221220   Samples5      65       182          1 0.246 0.364     BR
## 519 20221220   Samples5      66       182          1 0.194 0.292
## 520 20221220   Samples5      67    188spk          7 0.092 0.130
## 521 20221220   Samples5      68    188spk          7 0.093 0.126
## 522 20221220   Samples5      69    188spk          7 0.084 0.119
## 523 20221220   Samples5      70       196          7 0.200 0.299     BR
## 524 20221220   Samples5      71       196          7 0.150 0.227
## 525 20221220   Samples5      72       196          7 0.167 0.233
## 526 20221220   Samples5      73      Stds          1 0.037 0.041
## 527 20221220   Samples5      74      Stds          1 0.044 0.046
## 528 20221220   Samples5      75      Stds          1 0.038 0.040
## 529 20221220   Samples5      76       183          1 0.361 0.547
## 530 20221220   Samples5      77       183          1 0.353 0.516
## 531 20221220   Samples5      78       183          1 0.327 0.486     BR
## 532 20221220   Samples5      79       189          7 0.037 0.038
## 533 20221220   Samples5      80       189          7 0.037 0.039
## 534 20221220   Samples5      81       189          7 0.037 0.039
## 535 20221220   Samples5      82       197          7 0.138 0.188
## 536 20221220   Samples5      83       197          7 0.142 0.195
## 537 20221220   Samples5      84       197          7 0.129 0.180
## 538 20221220   Samples5      85      Stds          1 0.039 0.041
## 539 20221220   Samples5      86      Stds          1 0.039 0.041
## 540 20221220   Samples5      87      Stds          1 0.038 0.040
## 541 20221220   Samples5      88       184          1 1.036 1.234     BR
## 542 20221220   Samples5      89       184          1 0.852 1.133
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## 543 20221220   Samples5      90      184          1 0.886 1.229
## 544 20221220   Samples5      91      190          7 0.037 0.039
## 545 20221220   Samples5      92      190          7 0.038 0.040
## 546 20221220   Samples5      93      190          7 0.039 0.040
## 547 20221220   Samples5      94      198          7 0.154 0.245
## 548 20221220   Samples5      95      198          7 0.147 0.246
## 549 20221220   Samples5      96      198          7 0.157 0.218    BR
## 550 20221220   Samples6       1     Stds          1 1.306 1.425
## 551 20221220   Samples6       2     Stds          1 1.041 1.299
## 552 20221220   Samples6       3     Stds          1 1.120 1.364
## 553 20221220   Samples6       4      199          7 0.421 0.577
## 554 20221220   Samples6       5      199          7 0.422 0.618
## 555 20221220   Samples6       6      199          7 0.403 0.616
## 556 20221220   Samples6       7      207          7 0.338 0.510    BR
## 557 20221220   Samples6       8      207          7 0.284 0.388
## 558 20221220   Samples6       9      207          7 0.213 0.447
## 559 20221220   Samples6      10      213          7 0.689 0.892
## 560 20221220   Samples6      11      213          7 0.622 0.872
## 561 20221220   Samples6      12      213          7 0.433 0.668    BR
## 562 20221220   Samples6      13     Stds          1 0.054 0.070
## 563 20221220   Samples6      14     Stds          1 0.052 0.067
## 564 20221220   Samples6      15     Stds          1 0.052 0.069
## 565 20221220   Samples6      16      200          7 0.730 0.788    BR
## 566 20221220   Samples6      17      200          7 0.591 0.694
## 567 20221220   Samples6      18      200          7 0.624 0.700
## 568 20221220   Samples6      19      208          7 1.090 1.082
## 569 20221220   Samples6      20      208          7 0.943 0.987
## 570 20221220   Samples6      21      208          7 1.017 1.019
## 571 20221220   Samples6      22      214          7 0.558 0.610    BR
## 572 20221220   Samples6      23      214          7 0.782 0.854
## 573 20221220   Samples6      24      214          7 0.679 0.738
## 574 20221220   Samples6      25     Stds          1 0.039 0.040
## 575 20221220   Samples6      26     Stds          1 0.039 0.041
## 576 20221220   Samples6      27     Stds          1 0.038 0.040
## 577 20221220   Samples6      28      201          7 0.269 0.403    BR
## 578 20221220   Samples6      29      201          7 0.211 0.327
## 579 20221220   Samples6      30      201          7 0.224 0.371
## 580 20221220   Samples6      31      209          7 1.241 1.341
## 581 20221220   Samples6      32      209          7 1.319 1.369
## 582 20221220   Samples6      33      209          7 1.471 1.493
## 583 20221220   Samples6      34      215          7 1.492 1.471    ND
## 584 20221220   Samples6      35      215          7 1.580 1.612    ND
## 585 20221220   Samples6      36      215          7 1.352 1.464    ND
## 586 20221220   Samples6      37     Stds          1 0.037 0.038
## 587 20221220   Samples6      38     Stds          1 0.038 0.038
## 588 20221220   Samples6      39     Stds          1 0.037 0.038
## 589 20221220   Samples6      40      202          7 0.500 0.685    BR
## 590 20221220   Samples6      41      202          7 0.369 0.540
## 591 20221220   Samples6      42      202          7 0.443 0.572
## 592 20221220   Samples6      43     209d          7 1.381 1.477    BR
## 593 20221220   Samples6      44     209d          7 1.483 1.571
## 594 20221220   Samples6      45     209d          7 1.462 1.516
## 595 20221220   Samples6      46      216          1 0.051 0.048
## 596 20221220   Samples6      47      216          1 0.046 0.047
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## 597 20221220   Samples6      48       216          1 0.046 0.047
## 598 20221220   Samples6      49      Stds          1 0.042 0.167
## 599 20221220   Samples6      50      Stds          1 0.042 0.193
## 600 20221220   Samples6      51      Stds          1 0.042 0.201
## 601 20221220   Samples6      52       203          7 0.951 1.158    BR
## 602 20221220   Samples6      53       203          7 0.850 1.022
## 603 20221220   Samples6      54       203          7 0.788 1.161
## 604 20221220   Samples6      55       210          7 0.256 0.537
## 605 20221220   Samples6      56       210          7 0.262 0.349
## 606 20221220   Samples6      57       210          7 0.237 0.367
## 607 20221220   Samples6      58       217          1 0.063 0.065
## 608 20221220   Samples6      59       217          1 0.043 0.044
## 609 20221220   Samples6      60       217          1 0.043 0.045
## 610 20221220   Samples6      61      Stds          1 0.039 0.060
## 611 20221220   Samples6      62      Stds          1 0.046 0.065
## 612 20221220   Samples6      63      Stds          1 0.038 0.061
## 613 20221220   Samples6      64       204          7 0.360 0.528
## 614 20221220   Samples6      65       204          7 0.354 0.516
## 615 20221220   Samples6      66       204          7 0.346 0.472
## 616 20221220   Samples6      67    210spk          7 0.244 0.356    BR
## 617 20221220   Samples6      68    210spk          7 0.296 0.419
## 618 20221220   Samples6      69    210spk          7 0.267 0.402
## 619 20221220   Samples6      70       218          1 1.168 1.505
## 620 20221220   Samples6      71       218          1 1.457 1.757    BR
## 621 20221220   Samples6      72       218          1 1.299 1.635
## 622 20221220   Samples6      73      Stds          1 0.037 0.040
## 623 20221220   Samples6      74      Stds          1 0.037 0.039
## 624 20221220   Samples6      75      Stds          1 0.037 0.039
## 625 20221220   Samples6      76       205          7 0.842 1.118
## 626 20221220   Samples6      77       205          7 0.825 1.027
## 627 20221220   Samples6      78       205          7 0.731 0.999    BR
## 628 20221220   Samples6      79       211          7 0.673 0.867    BR
## 629 20221220   Samples6      80       211          7 0.823 1.069
## 630 20221220   Samples6      81       211          7 0.833 1.463
## 631 20221220   Samples6      82       219          1 0.050 0.051
## 632 20221220   Samples6      83       219          1 0.049 0.051
## 633 20221220   Samples6      84       219          1 0.049 0.051
## 634 20221220   Samples6      85      Stds          1 0.037 0.039
## 635 20221220   Samples6      86      Stds          1 0.037 0.038
## 636 20221220   Samples6      87      Stds          1 0.037 0.038
## 637 20221220   Samples6      88       206          7 1.883 1.820    ND
## 638 20221220   Samples6      89       206          7 1.865 1.954    ND
## 639 20221220   Samples6      90       206          7 1.985 1.945    ND
## 640 20221220   Samples6      91       212          7 1.779 1.841    ND
## 641 20221220   Samples6      92       212          7 1.739 1.779    ND
## 642 20221220   Samples6      93       212          7 1.856 1.788    ND
## 643 20221220   Samples6      94       220          1 0.044 0.045
## 644 20221220   Samples6      95       220          1 0.044 0.045
## 645 20221220   Samples6      96       220          1 0.043 0.046
## 646 20221220   Samples7       1      Stds          1 1.291 1.420
## 647 20221220   Samples7       2      Stds          1 1.137 1.373
## 648 20221220   Samples7       3      Stds          1 1.471 1.552
## 649 20221220   Samples7       4       221          1 0.050 0.051
## 650 20221220   Samples7       5       221          1 0.045 0.048
```

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## 651 20221220   Samples7     6     221          1 0.047 0.049
## 652 20221220   Samples7     7     229          7 0.250 0.512    BR
## 653 20221220   Samples7     8     229          7 0.239 0.355
## 654 20221220   Samples7     9     229          7 0.222 0.340
## 655 20221220   Samples7    10     235          7 0.179 0.255
## 656 20221220   Samples7    11     235          7 0.168 0.265
## 657 20221220   Samples7    12     235          7 0.166 0.253
## 658 20221220   Samples7    13    Stds          1 0.063 0.080
## 659 20221220   Samples7    14    Stds          1 0.060 0.075
## 660 20221220   Samples7    15    Stds          1 0.073 0.087
## 661 20221220   Samples7    16     222          1 0.052 0.053
## 662 20221220   Samples7    17     222          1 0.051 0.053
## 663 20221220   Samples7    18     222          1 0.051 0.053
## 664 20221220   Samples7    19     230          7 0.078 0.109
## 665 20221220   Samples7    20     230          7 0.073 0.093
## 666 20221220   Samples7    21     230          7 0.072 0.098
## 667 20221220   Samples7    22     236          7 0.077 0.090
## 668 20221220   Samples7    23     236          7 0.074 0.100
## 669 20221220   Samples7    24     236          7 0.077 0.099
## 670 20221220   Samples7    25    Stds          1 0.039 0.041
## 671 20221220   Samples7    26    Stds          1 0.039 0.041
## 672 20221220   Samples7    27    Stds          1 0.038 0.040
## 673 20221220   Samples7    28     223          1 0.045 0.048
## 674 20221220   Samples7    29     223          1 0.045 0.048
## 675 20221220   Samples7    30     223          1 0.045 0.047
## 676 20221220   Samples7    31     231          7 0.155 0.271
## 677 20221220   Samples7    32     231          7 0.156 0.230
## 678 20221220   Samples7    33     231          7 0.130 0.203
## 679 20221220   Samples7    34     237          7 0.046 0.050
## 680 20221220   Samples7    35     237          7 0.045 0.052
## 681 20221220   Samples7    36     237          7 0.046 0.050
## 682 20221220   Samples7    37    Stds          1 0.037 0.037
## 683 20221220   Samples7    38    Stds          1 0.037 0.038
## 684 20221220   Samples7    39    Stds          1 0.037 0.039
## 685 20221220   Samples7    40     224          1 0.045 0.047
## 686 20221220   Samples7    41     224          1 0.044 0.046
## 687 20221220   Samples7    42     224          1 0.044 0.045
## 688 20221220   Samples7    43    231d          7 0.149 0.239
## 689 20221220   Samples7    44    231d          7 0.155 0.246
## 690 20221220   Samples7    45    231d          7 0.139 0.235
## 691 20221220   Samples7    46     238          7 0.204 0.292
## 692 20221220   Samples7    47     238          7 0.204 0.319
## 693 20221220   Samples7    48     238          7 0.236 0.406    BR
## 694 20221220   Samples7    49    Stds          1 0.043 0.258
## 695 20221220   Samples7    50    Stds          1 0.042 0.236
## 696 20221220   Samples7    51    Stds          1 0.043 0.230
## 697 20221220   Samples7    52     225          1 0.047 0.048
## 698 20221220   Samples7    53     225          1 0.043 0.046
## 699 20221220   Samples7    54     225          1 0.044 0.045
## 700 20221220   Samples7    55     232          7 0.123 0.218
## 701 20221220   Samples7    56     232          7 0.124 0.234
## 702 20221220   Samples7    57     232          7 0.105 0.171    BR
## 703 20221220   Samples7    58     239          7 0.071 0.092
## 704 20221220   Samples7    59     239          7 0.066 0.095
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## 705 20221220   Samples7      60      239          7 0.083 0.122
## 706 20221220   Samples7      61     Stds          1 0.039 0.062
## 707 20221220   Samples7      62     Stds          1 0.039 0.062
## 708 20221220   Samples7      63     Stds          1 0.039 0.058
## 709 20221220   Samples7      64      226          7 0.058 0.069
## 710 20221220   Samples7      65      226          7 0.061 0.069
## 711 20221220   Samples7      66      226          7 0.055 0.066
## 712 20221220   Samples7      67   232spk          7 0.158 0.206
## 713 20221220   Samples7      68   232spk          7 0.139 0.201
## 714 20221220   Samples7      69   232spk          7 0.137 0.180
## 715 20221220   Samples7      70      240          7 0.293 0.385
## 716 20221220   Samples7      71      240          7 0.250 0.358
## 717 20221220   Samples7      72      240          7 0.244 0.370
## 718 20221220   Samples7      73     Stds          1 0.039 0.041
## 719 20221220   Samples7      74     Stds          1 0.037 0.040
## 720 20221220   Samples7      75     Stds          1 0.037 0.040
## 721 20221220   Samples7      76      227          7 0.056 0.067
## 722 20221220   Samples7      77      227          7 0.056 0.067
## 723 20221220   Samples7      78      227          7 0.050 0.058
## 724 20221220   Samples7      79      233          7 0.062 0.077
## 725 20221220   Samples7      80      233          7 0.057 0.074
## 726 20221220   Samples7      81      233          7 0.060 0.073
## 727 20221220   Samples7      82      241          7 0.132 0.162
## 728 20221220   Samples7      83      241          7 0.126 0.178
## 729 20221220   Samples7      84      241          7 0.130 0.168
## 730 20221220   Samples7      85     Stds          1 0.037 0.038
## 731 20221220   Samples7      86     Stds          1 0.037 0.038
## 732 20221220   Samples7      87     Stds          1 0.037 0.038
## 733 20221220   Samples7      88      228          7 0.055 0.082
## 734 20221220   Samples7      89      228          7 0.052 0.065
## 735 20221220   Samples7      90      228          7 0.054 0.067
## 736 20221220   Samples7      91      234          7 0.048 0.059
## 737 20221220   Samples7      92      234          7 0.047 0.056
## 738 20221220   Samples7      93      234          7 0.047 0.055
## 739 20221220   Samples7      94      242          7 0.070 0.143
## 740 20221220   Samples7      95      242          7 0.070 0.238
## 741 20221220   Samples7      96      242          7 0.072 0.249
## 742 20221220   Samples8       1     Stds          1 1.382 1.534
## 743 20221220   Samples8       2     Stds          1 1.477 1.530
## 744 20221220   Samples8       3     Stds          1 1.567 1.667
## 745 20221220   Samples8       4      243          7 0.280 0.512
## 746 20221220   Samples8       5      243          7 0.326 0.537
## 747 20221220   Samples8       6      243          7 0.300 0.529
## 748 20221220   Samples8       7      251          7 0.938 1.166
## 749 20221220   Samples8       8      251          7 0.861 1.003
## 750 20221220   Samples8       9      251          7 0.690 0.815    BR
## 751 20221220   Samples8      10      257          7 1.536 1.524    ND
## 752 20221220   Samples8      11      257          7 1.506 1.594    ND
## 753 20221220   Samples8      12      257          7 1.563 1.524    ND
## 754 20221220   Samples8      13     Stds          1 0.068 0.080
## 755 20221220   Samples8      14     Stds          1 0.063 0.077
## 756 20221220   Samples8      15     Stds          1 0.064 0.080
## 757 20221220   Samples8      16      244          7 0.450 0.586    BR
## 758 20221220   Samples8      17      244          7 0.360 0.512
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## 759 20221220   Samples8      18       244           7 0.362 0.502
## 760 20221220   Samples8      19       252           7 0.772 0.911      BR
## 761 20221220   Samples8      20       252           7 0.650 0.868
## 762 20221220   Samples8      21       252           7 0.704 0.832
## 763 20221220   Samples8      22       258           7 1.050 1.132
## 764 20221220   Samples8      23       258           7 1.250 1.258
## 765 20221220   Samples8      24       258           7 0.969 1.034      BR
## 766 20221220   Samples8      25      Stds           1 0.039 0.043
## 767 20221220   Samples8      26      Stds           1 0.039 0.040
## 768 20221220   Samples8      27      Stds           1 0.039 0.040
## 769 20221220   Samples8      28       245           7 0.807 1.008
## 770 20221220   Samples8      29       245           7 0.621 0.794      BR
## 771 20221220   Samples8      30       245           7 0.749 0.906
## 772 20221220   Samples8      31       253           7 0.811 1.028
## 773 20221220   Samples8      32       253           7 0.575 0.871      BR
## 774 20221220   Samples8      33       253           7 0.754 0.988
## 775 20221220   Samples8      34       259           7 1.478 1.549
## 776 20221220   Samples8      35       259           7 1.339 1.367
## 777 20221220   Samples8      36       259           7 1.190 1.283      BR
## 778 20221220   Samples8      37      Stds           1 0.037 0.038
## 779 20221220   Samples8      38      Stds           1 0.038 0.037
## 780 20221220   Samples8      39      Stds           1 0.037 0.038
## 781 20221220   Samples8      40       246           7 0.923 1.089      BR
## 782 20221220   Samples8      41       246           7 1.134 1.258
## 783 20221220   Samples8      42       246           7 1.048 1.223
## 784 20221220   Samples8      43      253d           7 0.638 0.827
## 785 20221220   Samples8      44      253d           7 0.504 0.649
## 786 20221220   Samples8      45      253d           7 0.752 1.014
## 787 20221220   Samples8      46       260           7 1.338 1.427
## 788 20221220   Samples8      47       260           7 1.316 1.385
## 789 20221220   Samples8      48       260           7 1.240 1.334      BR
## 790 20221220   Samples8      49      Stds           1 0.043 0.181
## 791 20221220   Samples8      50      Stds           1 0.044 0.163
## 792 20221220   Samples8      51      Stds           1 0.043 0.187
## 793 20221220   Samples8      52       247           7 1.227 1.349
## 794 20221220   Samples8      53       247           7 1.137 1.285
## 795 20221220   Samples8      54       247           7 0.983 1.159      BR
## 796 20221220   Samples8      55       254           7 0.111 0.175
## 797 20221220   Samples8      56       254           7 0.095 0.152
## 798 20221220   Samples8      57       254           7 0.117 0.182
## 799 20221220   Samples8      58       261           7 1.381 1.515      BR
## 800 20221220   Samples8      59       261           7 1.263 1.407
## 801 20221220   Samples8      60       261           7 1.310 1.443
## 802 20221220   Samples8      61      Stds           1 0.039 0.061
## 803 20221220   Samples8      62      Stds           1 0.039 0.060
## 804 20221220   Samples8      63      Stds           1 0.039 0.062
## 805 20221220   Samples8      64       248           7 0.896 1.058
## 806 20221220   Samples8      65       248           7 0.911 1.052
## 807 20221220   Samples8      66       248           7 0.882 1.041
## 808 20221220   Samples8      67    254spk           7 0.143 0.210
## 809 20221220   Samples8      68    254spk           7 0.137 0.193
## 810 20221220   Samples8      69    254spk           7 0.153 0.214
## 811 20221220   Samples8      70       262           7 1.020 1.083      BR
## 812 20221220   Samples8      71       262           7 1.260 1.343
```

```
## 813 20221220   Samples8      72      262          7 1.260 1.339
## 814 20221220   Samples8      73     Stds          1 0.038 0.039
## 815 20221220   Samples8      74     Stds          1 0.037 0.040
## 816 20221220   Samples8      75     Stds          1 0.038 0.039
## 817 20221220   Samples8      76      249          7 0.910 1.105
## 818 20221220   Samples8      77      249          7 0.965 1.129
## 819 20221220   Samples8      78      249          7 0.816 1.013    BR
## 820 20221220   Samples8      79      255          7 0.089 0.165
## 821 20221220   Samples8      80      255          7 0.089 0.140
## 822 20221220   Samples8      81      255          7 0.086 0.146
## 823 20221220   Samples8      82      263          7 1.076 1.138
## 824 20221220   Samples8      83      263          7 1.023 1.158
## 825 20221220   Samples8      84      263          7 1.167 1.277    BR
## 826 20221220   Samples8      85     Stds          1 0.037 0.037
## 827 20221220   Samples8      86     Stds          1 0.037 0.038
## 828 20221220   Samples8      87     Stds          1 0.037 0.038
## 829 20221220   Samples8      88      250          7 0.633 0.815
## 830 20221220   Samples8      89      250          7 0.853 1.075    BR
## 831 20221220   Samples8      90      250          7 0.699 0.955
## 832 20221220   Samples8      91      256          7 1.072 1.241
## 833 20221220   Samples8      92      256          7 0.039 0.079    BR
## 834 20221220   Samples8      93      256          7 1.004 1.178
## 835 20221220   Samples8      94      264          7 1.355 1.426
## 836 20221220   Samples8      95      264          7 1.417 1.450
## 837 20221220   Samples8      96      264          7 0.195 0.227    BR
```

```
dat <- dat %>%  filter(!Notes =='BR')
dat <- dat %>%  filter(!Notes =='ND')
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400
## 4 20221220 Dilutions2       4       46          7 1.196 1.303
## 5 20221220 Dilutions2       5       46          7 1.030 1.338
## 6 20221220 Dilutions2       7       62          7 0.987 1.182
```

```
#Now flag any Abs1 that are above the 60uM standard absorbance (0.2)
dat$FE_2_Curve <- ifelse(dat$Abs1 >0.2, "High", "Low")
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
```

```
#Now flag any Abs2 that are above the 60uM standard absorbance (0.3)
dat$FE_Tot_Curve <- ifelse(dat$Abs2 >0.3, "High", "Low")
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve
## 1         High
## 2         High
## 3         High
## 4         High
## 5         High
## 6         High
```

Subset Data and Calculate Concentrations

```
#Calculate concentrations of Fe(II)
dat$FE_2_Conc <- ifelse(dat$Abs1 >0.2, (dat$Abs1-Slopes2$Intercept)/Slopes2$Slope,
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc
## 1         High  429.1814
## 2         High  470.6580
## 3         High  412.0814
## 4         High  418.9942
## 5         High  358.5985
## 6         High  342.9538
```

```
#calculate concentrations of Fe(total)
dat$FE_Tot_Conc <- ifelse(dat$Abs2 >0.3, (dat$Abs2-Slopes4$Intercept)/Slopes4$Slope,
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc
## 1         High  429.1814    438.8547
## 2         High  470.6580    451.6973
## 3         High  412.0814    432.2687
```

```
## 4         High  418.9942    400.3268
## 5         High  358.5985    411.8522
## 6         High  342.9538    360.4817
```

```
#Account for the dilution factor
dat$FE_2_Conc_DilCorr <- dat$FE_2_Conc*dat$Dil_Factor

dat$FE_Tot_Conc_DilCorr <- dat$FE_Tot_Conc*dat$Dil_Factor

head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1         High  429.1814    438.8547          429.1814            438.8547
## 2         High  470.6580    451.6973          470.6580            451.6973
## 3         High  412.0814    432.2687          412.0814            432.2687
## 4         High  418.9942    400.3268         2932.9593           2802.2874
## 5         High  358.5985    411.8522         2510.1893           2882.9655
## 6         High  342.9538    360.4817         2400.6765           2523.3717
```

Calculate Fe(III)

```
#Calculate concentrations of Fe(III)
dat$FE_3_Conc <- dat$FE_Tot_Conc_DilCorr-dat$FE_2_Conc_DilCorr

#Use ifelse to make any negative values equal to zero
dat$FE_2_Conc_Final <- ifelse(dat$FE_2_Conc_DilCorr <0, 0, dat$FE_2_Conc_DilCorr)

dat$FE_3_Conc_Final <- ifelse(dat$FE_3_Conc <0, 0, dat$FE_3_Conc)

#Use ifelse to make any negative values equal to zero
dat$FE_2_Info <- ifelse(dat$FE_2_Conc_DilCorr <0, 'bdl', 'within range')

#Use ifelse to make any negative values equal to zero
dat$FE_3_Info <- ifelse(dat$FE_3_Conc <0, 'bdl', 'within range')

head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
```

```
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1         High  429.1814    438.8547          429.1814            438.8547
## 2         High  470.6580    451.6973          470.6580            451.6973
## 3         High  412.0814    432.2687          412.0814            432.2687
## 4         High  418.9942    400.3268         2932.9593           2802.2874
## 5         High  358.5985    411.8522         2510.1893           2882.9655
## 6         High  342.9538    360.4817         2400.6765           2523.3717
##     FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final    FE_2_Info    FE_3_Info
## 1    9.673267        429.1814        9.673267 within range within range
## 2  -18.960674        470.6580        0.000000 within range          bdl
## 3   20.187292        412.0814       20.187292 within range within range
## 4 -130.671902       2932.9593        0.000000 within range          bdl
## 5  372.776238       2510.1893      372.776238 within range within range
## 6  122.695121       2400.6765      122.695121 within range within range
```

Calculate Averages across wells and std. dev.

```r
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1         High  429.1814    438.8547          429.1814            438.8547
## 2         High  470.6580    451.6973          470.6580            451.6973
## 3         High  412.0814    432.2687          412.0814            432.2687
## 4         High  418.9942    400.3268         2932.9593           2802.2874
## 5         High  358.5985    411.8522         2510.1893           2882.9655
## 6         High  342.9538    360.4817         2400.6765           2523.3717
##     FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final    FE_2_Info    FE_3_Info
## 1    9.673267        429.1814        9.673267 within range within range
## 2  -18.960674        470.6580        0.000000 within range          bdl
## 3   20.187292        412.0814       20.187292 within range within range
## 4 -130.671902       2932.9593        0.000000 within range          bdl
## 5  372.776238       2510.1893      372.776238 within range within range
## 6  122.695121       2400.6765      122.695121 within range within range
```

```r
#summarize by sampleID so that we can calculate the mean and std. dev. of the three wells
dat1 <- dat %>%
  group_by(SampleID) %>%
  summarise(FE_2_mean = mean(FE_2_Conc_Final), FE_2_sd = sd(FE_2_Conc_Final),
            FE_3_mean = mean(FE_3_Conc_Final), FE_3_sd = sd(FE_3_Conc_Final),
            FE_2_flag = first(FE_2_Info), FE_3_flag = first(FE_3_Info))

head(dat1)
```

```
## # A tibble: 6 x 7
```

```
##   SampleID FE_2_mean FE_2_sd FE_3_mean FE_3_sd FE_2_flag    FE_3_flag
##   <chr>        <dbl>   <dbl>     <dbl>   <dbl> <chr>        <chr>
## 1 10           63.7    6.77     279.     26.2  within range within range
## 2 100          14.5    0.659      0       0    within range bdl
## 3 100spk       41.0    1.12       5.20    4.87 within range within range
## 4 101           8.26   0.317      0       0    within range bdl
## 5 102           9.63   0.183      0       0    within range bdl
## 6 103           9.42   0.659      0       0    within range bdl
```

Export full data then just final data

```r
#Read in meta-data and sample IDs
IDs <- read.csv("Raw Data/FE_SampleIDs_2022.csv")
head(IDs)
```

```
##       Date Month Year Campaign Site Zone Depth Rep
## 1 20220502   May 2022   MonMon  MSM   WC    10   A
## 2 20220502   May 2022   MonMon  MSM   WC    10   B
## 3 20220502   May 2022   MonMon  MSM   WC    10   C
## 4 20220502   May 2022   MonMon  MSM   WC    20   A
## 5 20220502   May 2022   MonMon  MSM   WC    20   B
## 6 20220502   May 2022   MonMon  MSM   WC    20   C
##                            Sample_Name SampleID
## 1 MSM_WC_PW_MonMon_20220502_SipA_10cm      654
## 2 MSM_WC_PW_MonMon_20220502_SipB_10cm      656
## 3 MSM_WC_PW_MonMon_20220502_SipC_10cm      658
## 4 MSM_WC_PW_MonMon_20220502_SipA_20cm      655
## 5 MSM_WC_PW_MonMon_20220502_SipB_20cm      657
## 6 MSM_WC_PW_MonMon_20220502_SipC_20cm      659
```

```r
dat_IDs <- merge(IDs, dat1)
head(dat_IDs)
```

```
##   SampleID   Date    Month Year Campaign  Site Zone Depth Rep
## 1        5 202211 November 2022   MonMon Gcrew   TR    20   A
## 2        8 202211 November 2022   MonMon Gcrew   TR    20   B
## 3        9 202211 November 2022   MonMon Gcrew   TR    45   B
## 4       10 202211 November 2022   MonMon Gcrew   TR    10   C
## 5       11 202211 November 2022   MonMon Gcrew   TR    20   C
## 6       12 202211 November 2022   MonMon Gcrew   TR    45   C
##          Sample_Name  FE_2_mean   FE_2_sd FE_3_mean   FE_3_sd    FE_2_flag
## 1 GCrew_TR_LysA_20cm  570.79540 27.012972  176.5054  43.07461 within range
## 2 GCrew_TR_LysB_20cm  631.49431 47.121673  644.8244 104.08377 within range
## 3 GCrew_TR_LysB_45cm 1022.85372 57.627674    0.0000   0.00000 within range
## 4 GCrew_TR_LysC_10cm   63.74255  6.771533  278.9614  26.17102 within range
## 5 GCrew_TR_LysC_20cm  307.18863 10.971138  447.0275  32.16041 within range
## 6 GCrew_TR_LysC_45cm  586.07625 16.207783  540.4117  45.54677 within range
##      FE_3_flag
## 1 within range
## 2 within range
## 3          bdl
```

```
## 4 within range
## 5 within range
## 6 within range
```

```
#Read out the summarized data
head(dat_IDs)
```

```
##   SampleID   Date    Month Year Campaign  Site Zone Depth Rep
## 1        5 202211 November 2022   MonMon Gcrew   TR    20   A
## 2        8 202211 November 2022   MonMon Gcrew   TR    20   B
## 3        9 202211 November 2022   MonMon Gcrew   TR    45   B
## 4       10 202211 November 2022   MonMon Gcrew   TR    10   C
## 5       11 202211 November 2022   MonMon Gcrew   TR    20   C
## 6       12 202211 November 2022   MonMon Gcrew   TR    45   C
##          Sample_Name  FE_2_mean   FE_2_sd FE_3_mean   FE_3_sd    FE_2_flag
## 1 GCrew_TR_LysA_20cm  570.79540 27.012972  176.5054  43.07461 within range
## 2 GCrew_TR_LysB_20cm  631.49431 47.121673  644.8244 104.08377 within range
## 3 GCrew_TR_LysB_45cm 1022.85372 57.627674    0.0000   0.00000 within range
## 4 GCrew_TR_LysC_10cm   63.74255  6.771533  278.9614  26.17102 within range
## 5 GCrew_TR_LysC_20cm  307.18863 10.971138  447.0275  32.16041 within range
## 6 GCrew_TR_LysC_45cm  586.07625 16.207783  540.4117  45.54677 within range
##      FE_3_flag
## 1 within range
## 2 within range
## 3          bdl
## 4 within range
## 5 within range
## 6 within range
```

```
write.csv(dat_IDs, file="Processed Data/FE_20221220_Summary_Data.csv")

#Read out all the data in dat
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1         High  429.1814    438.8547          429.1814            438.8547
## 2         High  470.6580    451.6973          470.6580            451.6973
## 3         High  412.0814    432.2687          412.0814            432.2687
## 4         High  418.9942    400.3268         2932.9593           2802.2874
## 5         High  358.5985    411.8522         2510.1893           2882.9655
## 6         High  342.9538    360.4817         2400.6765           2523.3717
##     FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final    FE_2_Info    FE_3_Info
## 1    9.673267        429.1814        9.673267 within range within range
## 2  -18.960674        470.6580        0.000000 within range          bdl
## 3   20.187292        412.0814       20.187292 within range within range
## 4 -130.671902       2932.9593        0.000000 within range          bdl
```

```
## 5  372.776238       2510.1893      372.776238 within range within range
## 6  122.695121       2400.6765      122.695121 within range within range
```

```r
write.csv(dat, file="Processed Data/FE_20221220_Data.csv")


#Now take out the absorbance and stuff
head(dat)
```

```
##       Date      Plate Well_ID SampleID Dil_Factor  Abs1  Abs2 Notes FE_2_Curve
## 1 20221220 Dilutions2       1     stds          1 1.224 1.420             High
## 2 20221220 Dilutions2       2     stds          1 1.338 1.459             High
## 3 20221220 Dilutions2       3     stds          1 1.177 1.400             High
## 4 20221220 Dilutions2       4       46          7 1.196 1.303             High
## 5 20221220 Dilutions2       5       46          7 1.030 1.338             High
## 6 20221220 Dilutions2       7       62          7 0.987 1.182             High
##   FE_Tot_Curve FE_2_Conc FE_Tot_Conc FE_2_Conc_DilCorr FE_Tot_Conc_DilCorr
## 1         High  429.1814    438.8547          429.1814            438.8547
## 2         High  470.6580    451.6973          470.6580            451.6973
## 3         High  412.0814    432.2687          412.0814            432.2687
## 4         High  418.9942    400.3268         2932.9593           2802.2874
## 5         High  358.5985    411.8522         2510.1893           2882.9655
## 6         High  342.9538    360.4817         2400.6765           2523.3717
##     FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final    FE_2_Info    FE_3_Info
## 1    9.673267        429.1814        9.673267 within range within range
## 2  -18.960674        470.6580        0.000000 within range          bdl
## 3   20.187292        412.0814       20.187292 within range within range
## 4 -130.671902       2932.9593        0.000000 within range          bdl
## 5  372.776238       2510.1893      372.776238 within range within range
## 6  122.695121       2400.6765      122.695121 within range within range
```

```r
dat2 <- dat[ ,-(2:3)]
dat2 <- dat2[ ,-(3:12)]
head(dat2)
```

```
##       Date SampleID   FE_3_Conc FE_2_Conc_Final FE_3_Conc_Final    FE_2_Info
## 1 20221220     stds    9.673267        429.1814        9.673267 within range
## 2 20221220     stds  -18.960674        470.6580        0.000000 within range
## 3 20221220     stds   20.187292        412.0814       20.187292 within range
## 4 20221220       46 -130.671902       2932.9593        0.000000 within range
## 5 20221220       46  372.776238       2510.1893      372.776238 within range
## 6 20221220       62  122.695121       2400.6765      122.695121 within range
##      FE_3_Info
## 1 within range
## 2          bdl
## 3 within range
## 4          bdl
## 5 within range
## 6 within range
```

```r
write.csv(dat2, file="Processed Data/FE_20221220_short_Data.csv")
```

END